

Name: NEPTUN code: Major:

Probability Theory 1 exam, 13th January 2026

Working hours: 100 min. A non-programmable calculator without an internet connection can be used.

Maximal amount of points (with the Bonus): 110 points, but 100 points are already considered as 100%.

The. 1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

- (a) (2 points) Define the conditional probability with respect to an event E .
- (b) (2 points) Give the definition of complete event systems (partitions)!
- (c) (2+3 points) State and prove the theorem of complete event systems (also known as the law of total probability).
- (d) (2+3 points) State and prove Bayes' Theorem.

The. 2. (a) (2 points) Give the probability density function (p.d.f.) of a random variable with normal distribution $N(\mu, \sigma^2)$.

- (b) (7 points) Prove that it is indeed a p.d.f. *Hint: why is it enough to prove it in the standardized case?*
- (c) (4 points) Give the joint probability density function of the random variables (X, Y) with joint normal distribution.
- (d) (7 points) Suppose that $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ and X and Y are independent. Give the distribution of $X + Y$, and verify it. *Hint: why is it enough to prove it in the standardized case?*

The. 3. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let X and Y be random variables.

- (a) (2+2 points) Define the covariance $\text{cov}(X, Y)$ and the correlation $\text{corr}(X, Y)$ of X and Y .
- (b) (2+6 points) State and prove the Cauchy-Schwartz inequality.
- (c) (4 points) What can we say about the relation of X and Y if $\text{corr}(X, Y) = -1$? Verify your statement!

Prac. 1. The arrival times of the trams at the tram stops of Randomopolis follow a Poisson process. The Poisson processes of different tram lines are independent. On average, there are 12 trams per hour at a tram stop.

- (a) (5 points) I arrived to the stop at 8:00 a.m. What is the probability that I have to wait for a tram for more than 6 minutes?
- (b) (8 points) I will have an appointment at 8:40. To get there, I have to commute once. I take the first tram that arrives and travel with it for 15 minutes, when I get off. Then, at the stop where I got off, I waited for another tram and travelled with it for 10 minutes. What is the probability that I arrive at my appointment on time?
- (c) (5 points) On each of the trams that arrive at a particular stop, there are always 30 passengers, and each of them takes off independently with probability $1/3$. What is the expected value of the passengers taking off at the stop between 8:00 and 10:00?

Bonus: (10 points) What is the variance of the number of passengers taking off at the stop between 8:00 and 10:00?

Prac. 2. (16 points) I have two coins: an unbiased one and a biased one, which lands on heads with probability $3/5$. I forgot which is which. I picked one at random and tossed it 100 times; it landed heads 58 times. Estimate the probability that I picked the unbiased one using de Moivre-Laplace.

Prac. 3. Take a PQ line segment of length 2 and break it into two parts at a uniformly chosen point R . Let X denote the length of the line segment PR . We broke the other RQ segment again at a uniformly chosen point S . Let Y denote the length of the line segment RS .

- (a) (6 points) What is the joint probability distribution function of (X, Y) ?
- (b) (4 points) Find $\mathbb{E}(Y|X = x)$.
- (c) (6 points) Calculate $\text{cov}(X, Y)$.

	$\Phi(z)$									
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000